

Ex.No. 11: Implementation of IoT-based Smart Agriculture using MATLAB

Objective 1 Matlab Code and Output:

```
clc; clear; close all;

% Simulated soil moisture data (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold for irrigation

threshold = 40;

figure

% ----- Graph 1: Soil Moisture Variation -----

subplot(2,1,1)

plot(time, soil_moisture, '-o', 'LineWidth', 2)

title('Soil Moisture Monitoring in IoT Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

ylines(threshold,'r--','Threshold')

% ----- Graph 2: Moisture Status (Dry/Wet Level) -----

status = soil_moisture >= threshold;

subplot(2,1,2)

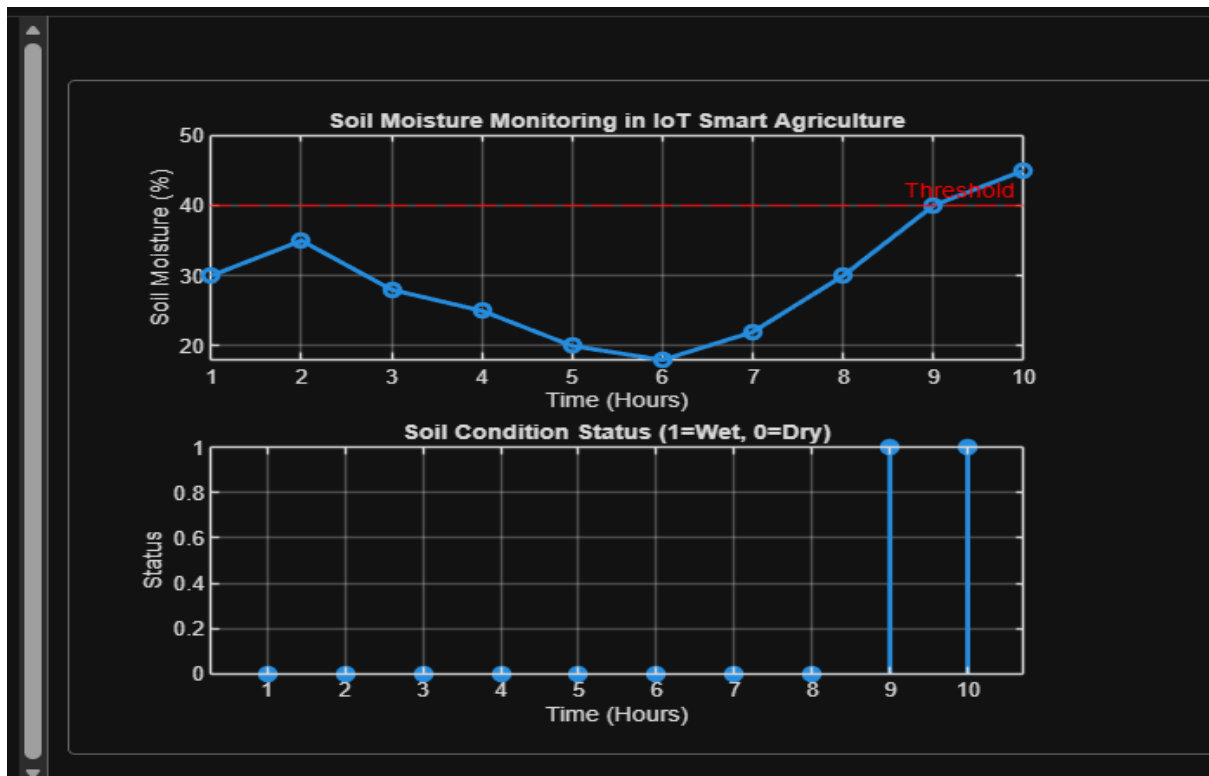
stem(time, status, 'filled', 'LineWidth', 2)

title('Soil Condition Status (1=Wet, 0=Dry)')

xlabel('Time (Hours)')

ylabel('Status')

grid on
```



Objective 2 Matlab Code and Output:

```
clc; clear; close all;

% Simulated soil moisture (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level

threshold = 40;

% Pump control logic (1 = ON, 0 = OFF)

pump_status = soil_moisture < threshold;

figure

% ----- Graph 1: Soil Moisture vs Pump Status -----

subplot(2,1,1)

plot(time, soil_moisture, '-o','LineWidth',2)

title('Soil Moisture vs Pump Control')

xlabel('Time (Hours)')
```

```

ylabel('Soil Moisture (%)')

grid on

hold on

yline(threshold,'r--','Threshold')

% ----- Graph 2: Pump Status (ON/OFF) -----

subplot(2,1,2)

stem(time, pump_status, 'filled','LineWidth',2)

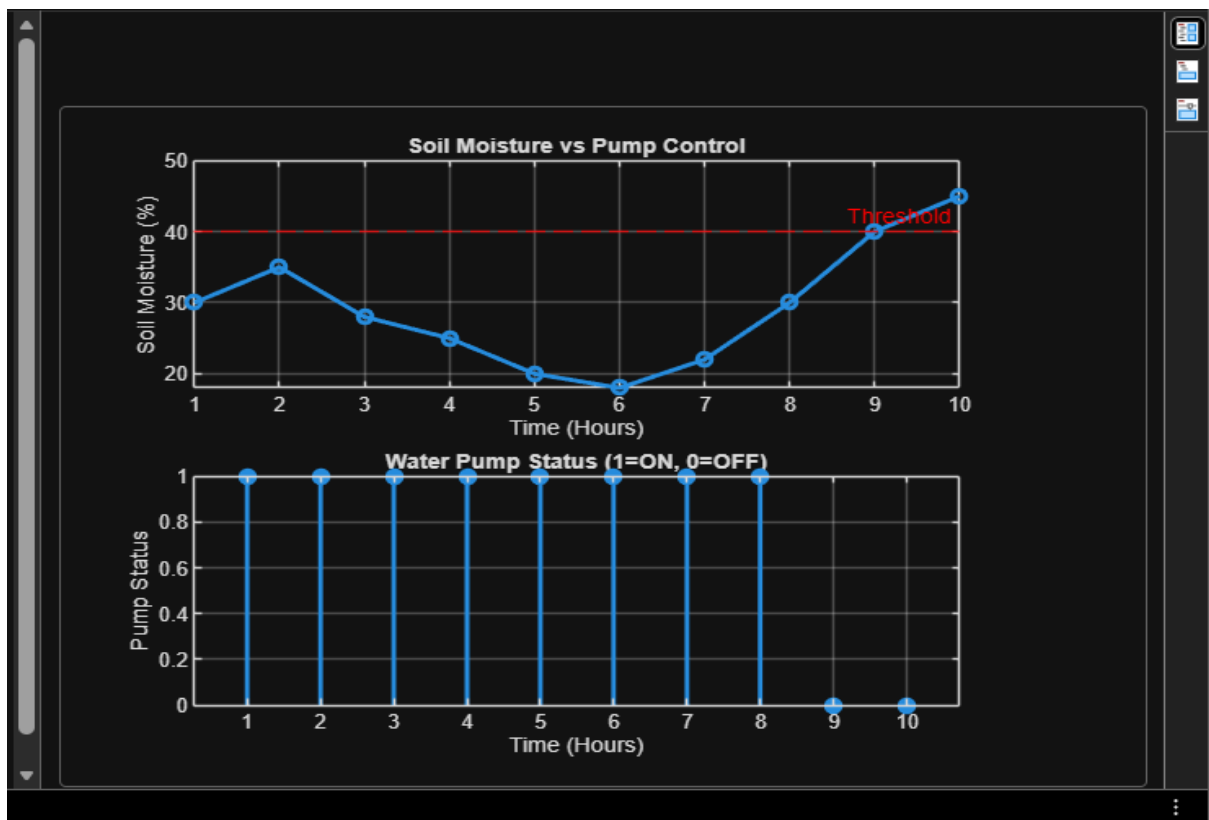
title('Water Pump Status (1=ON, 0=OFF)')

xlabel('Time (Hours)')

ylabel('Pump Status')

grid on

```



Objective 3 Matlab Code and Output:

```
clc; clear; close all;

% Simulated soil moisture data (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level

threshold = 40;

% Irrigation efficiency calculation (%)

efficiency = (soil_moisture ./ threshold) * 100;

efficiency(efficiency > 100) = 100; % limit to 100%

figure

% ----- Graph 1: Soil Moisture vs Efficiency -----

subplot(2,1,1)

plot(time, soil_moisture, '-o','LineWidth',2)

title('Soil Moisture Variation')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

yline(threshold,'r--','Threshold')

% ----- Graph 2: Irrigation Efficiency -----

subplot(2,1,2)

plot(time, efficiency, '-s','LineWidth',2)

title('Irrigation Efficiency (%) in Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Efficiency (%)')

grid on
```

